CliniCause: An End-to-End Framework for Constructing Clinically Grounded Causal Analysis Datasets

Anonymous submission

Abstract

1 Introduction

Causal inference asks how outcomes would change under alternative actions, rather than only which outcomes can be predicted. This distinction is central to medicine and public policy: an accurate forecast may identify who is at risk, but it does not establish which intervention would alter that risk. Modern causal machine-learning methods support flexible adjustment and heterogeneous-effect estimation (Chernozhukov et al. 2018; Wager and Athey 2018; Shalit, Johansson, and Sontag 2017; Louizos et al. 2017), but evaluating them is unusually difficult. For each individual, the counterfactual outcome needed to score an estimated effect is unobserved.

Because observational data do not reveal counterfactual outcomes, they do not provide a direct answer key for evaluating causal estimators. They also leave the true treatment-assignment process and the extent of unmeasured confounding unknown. The safest experimental sandbox is therefore fully synthetic data: the evaluator specifies the covariates, treatment assignment, potential outcomes, confounding, overlap, and effect heterogeneity. Assumptions such as ignorability or positivity can be satisfied, violated, or gradually stressed under complete control, while the true causal effect remains known.

That control creates an important limitation. The evaluator determines not only the answer but also the difficulty of recovering it. A method may perform well because its assumptions align with the selected assignment model, response surface, confounding structure, or overlap regime. Semi-synthetic benchmarks move closer to observed data by retaining empirical covariates while simulating treatment assignments, potential outcomes, or both. RCT-derived and empirically anchored designs preserve still more of the observed system, but continue to obtain an answer key through researcher-controlled sampling, comparison, or generation mechanisms. Across this spectrum, greater control makes accuracy measurable, while greater observational fidelity makes the causal target less knowable.

Several widely used causal-ML resources illustrate this spectrum of constructed evaluation. IHDP retains empirical covariates while simulating outcomes, whereas ACIC uses empirical covariates within diverse simulated treatment and outcome mechanisms; Dragonnet, for example, evaluates 101 retained datasets drawn from 63 ACIC 2018 data-generating settings (Shalit, Johansson, and Sontag 2017; Shi, Blei, and Veitch 2019). Influence-function validation similarly compares models across 77 semi-synthetic competition datasets sharing one empirical feature source but using distinct simulated assignments and outcomes (Alaa and van der Schaar 2019). More empirically grounded designs retain additional observed components: observational sampling from randomized trials preserves real trial variables but induces confounding through researcher-specified selection (Gentzel, Pruthi, and Jensen 2021); Jobs/LaLonde combines experimental treated subjects with external nonexperimental controls, making comparability depend on population-selection and harmonization choices; and Credence learns from an empirical distribution but generates data under user-specified effects and confounding (Parikh et al. 2022). These designs are increasingly sophisticated and provide valuable causal answer keys. Nevertheless, measurable accuracy still depends on an experimental anchor or a researcher-constructed causal, sampling, or comparison mechanism. They therefore cannot by themselves show how estimators behave when exposure prevalence, missingness, support, and measurement processes arise from routine practice rather than from the evaluation design.

In this paper, we propose a pivot in where researchers intervene when constructing causal-evaluation resources. Before defining this paradigm formally, consider a running ICU example in which the goal is to estimate how a shock state affects in-hospital mortality. The raw record does not contain a single, ready-made shock variable. Instead, it contains an irregular sequence of blood-pressure measurements, lactate tests, urine-output observations, and vasopressor administrations, followed by a source-observed mortality outcome.

A fully synthetic benchmark could generate the patients, assign shock according to a researcher-specified probability, and simulate mortality under shock and no shock. A semi-synthetic benchmark could retain observed patient covariates while simulating the shock assignment, mortality outcomes, or both. Either construction provides a known causal answer, but both the answer and the difficulty of recovering it depend on researcher-selected mechanisms.

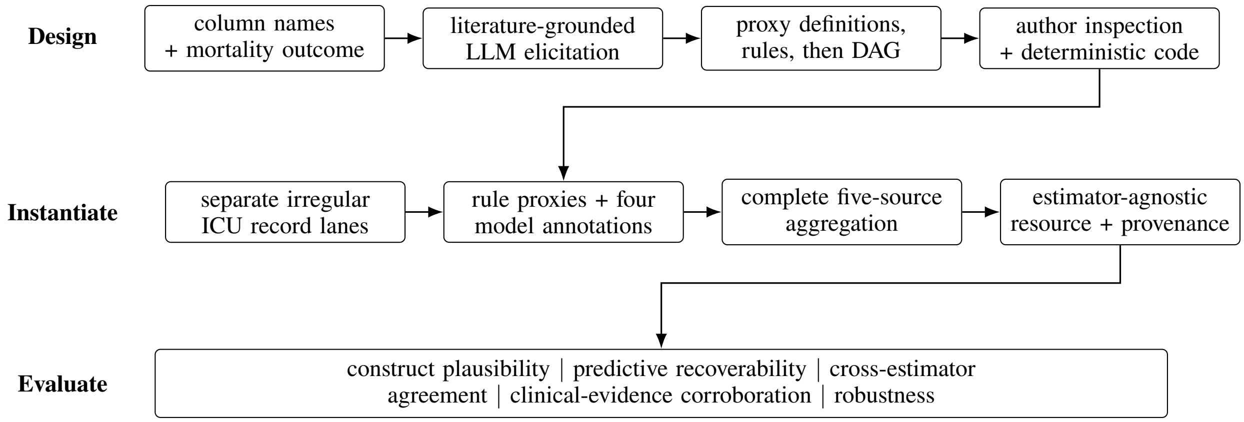

Figure 1: CliniCause separates knowledge-driven design from patient-level instantiation and empirical evaluation. The LLM receives schema information, not patient rows; deterministic code applies the accepted representation in separate dataset lanes. Evaluation is downstream of resource construction.

We propose shifting this intervention from the data-generating process to the *representation layer*. In the running example, the evaluator preserves the shock-related measurements, clinical practice patterns, missingness, empirical support, and mortality recorded during routine care. The evaluator intervenes by specifying which measurements and temporal rules define the shock exposure, which patients enter the cohort, which variables serve as covariates and outcomes, and which causal relationships determine adjustment.

Crucially, these choices are knowledge-driven rather than learned from exposure–outcome associations in the evaluation data. Clinical experts, published evidence, or externally elicited domain knowledge specify, for example, that hypotension, elevated lactate, low urine output, and vasopressor use constitute evidence of shock; the dataset is not searched for the definition that produces the strongest mortality effect. Patient records are used to instantiate this prespecified representation. This separation reduces outcome leakage and post-hoc selection while making every threshold, temporal rule, cohort restriction, and graph edge an explicit knowledge claim that can be inspected, challenged, and replaced.

LLMs make this knowledge-driven construction increasingly scalable. Their broad clinical knowledge can support structured elicitation of candidate constructs and relationships (Singhal et al. 2023), while their reliability limitations require that every proposal remain auditable and subordinate to external evidence and expert review (Darvariu, Hailes, and Musolesi 2024). The LLM therefore serves as a design assistant rather than a source of causal ground truth: it helps make the proposed representation explicit but does not process patient records, generate exposures or outcomes, or select definitions according to their estimated effects.

The running example reflects the real setting studied in this paper. We instantiate the representation-centered paradigm through *CliniCause*, an evidence-tracked pipeline applied to longitudinal ICU records from MIMIC-III and PhysioNet 2012 (Johnson et al. 2016; Silva et al. 2012).

CliniCause uses ChatGPT through its desktop interface as a knowledge-elicitation assistant at the representation stage. ChatGPT was given the names of the columns available in the ICU time series and the target outcome, but not the patient-level records, variable distributions, exposure prevalences, causal estimates, or access to the environment containing MIMIC-III and PhysioNet. Drawing on clinical literature available through its pretrained knowledge and online search, it was prompted to propose clinically meaningful categorical proxy exposures, the measurement cutoffs and temporal rules needed to instantiate them, and a DAG relating the proposed constructs, observed covariates, and mortality. Every proposed proxy, cutoff, and DAG edge had to be accompanied by a clinical rationale and a supporting literature source, making each knowledge claim traceable and open to verification.

Dan: Kobi — we need a separate supplementary PDF documenting the clinical evidence used to construct the CliniCause representation. Please prepare a LaTeX file containing two tables:
1. Proxy-definition table: one row for every proxy exposure in both MIMIC-III and PhysioNet. Include the proxy name, source columns, exact cutoff(s), temporal/aggregation rule, clinical interpretation, and the paper(s) that ChatGPT supplied as support. Please include a short paraphrase of what each source supports and its DOI/URL.
2. DAG-edge table: one row for every edge in each dataset-specific graph. Include the parent node, child node, clinical rationale, supporting paper(s) supplied by ChatGPT, and whether you independently verified that the cited paper actually supports that relationship. .

The authors inspected the proposed rationales and translated the selected proxies and relationships into explicit rules and graph assumptions, yielding states such as renal dysfunction, hepatic dysfunction, shock, respiratory failure, cardiac strain, and sepsis-related burden. Because the LLM had neither patient-level inputs nor access to the data environment, the exposure definitions could not be optimized against observed MIMIC-III or PhysioNet outcomes during elicitation.

After construction was complete, we conducted a separate literature-based corroboration of the resulting effect directions (Table 4); this post-construction audit was not used to select the proxies, tune their thresholds, or construct the DAG.

The resulting resources contain 26,845 MIMIC-III and 7,993 PhysioNet patient records, with nine and ten admitted proxy exposures, respectively. Rather than merely compressing the underlying time series, each resource provides an explicit causal-analysis interface connecting source-observed measurements and mortality to candidate exposures, covariates, cohort rules, graph assumptions, adjustment choices, and provenance.

By preserving the naturally occurring measurement, exposure, support, and outcome processes, CliniCause gives up the known answer provided by controlled benchmarks. Its credibility must therefore come from converging evidence rather than comparison with causal ground truth. We examine the resources along four complementary axes: whether the candidate exposures and graph assumptions are clinically plausible; whether the rule-derived constructs are recoverable from the underlying measurements; whether distinct causal estimators agree once the representation is fixed; and whether the estimated directions corroborate mortality relationships reported in independent clinical studies. Together, these checks assess whether CliniCause provides a convincing and useful observational causal testbed, without treating its proxies as literal interventions, its DAG as known to be correct, or its estimated effects as ground-truth causal answers.

We evaluate the resulting resources along each of the four proposed axes. Informal consultations with ICU physicians provided formative feedback on the clinical plausibility of the proxies and graph assumptions, while predictive experiments assessed whether the rule-derived proxies could be recovered from the underlying measurements. Across 19 prespecified dataset–exposure comparisons, CausalForestDML and LinearDML agreed in direction in all 19, and CausalPFN agreed in 18; PhysioNet shock was the sole exception. Finally, comparison with published mortality gradients supported the renal, hepatic, cardiac, inflammation/sepsis, and global-severity directions, while identifying shock and PhysioNet coagulation/hematologic dysfunction as clinically discordant cases requiring further review (Table 4). The physician consultations were informal rather than structured clinical adjudication, and the combined checks provide converging evidence rather than causal ground truth.

Our contributions are:

1. an evidence-tracked pipeline that converts irregular clinical records into explicit causal-analysis representations while preserving observed patient, support, and outcome processes;
2. two reusable resources derived from MIMIC-III and PhysioNet 2012, containing 19 clinically motivated proxy-exposure analyses with inspectable cohort, graph, adjustment, and provenance assumptions; and
3. an evaluation framework for observational causal resources based on construct plausibility, proxy recoverability, cross-estimator agreement, and clinical-literature corroboration, including explicit identification of discordant results.

2 Related Work

Clinical phenotyping and temporal abstraction. Electronic phenotyping converts heterogeneous health records into computable clinical constructs using expert rules, learned models, or combinations of labeling sources (Banda et al. 2018; Ratner et al. 2020). For irregular clinical time series, GRU-D models informative missingness, whereas STraTS represents observations as sparse time–variable–value events (Che et al. 2018; Tipirneni and Reddy 2022). These methods recover or predict patient-level states from observed data. CliniCause instead uses externally elicited clinical knowledge to determine which proxy states should exist and how they should be operationalized, then uses temporal models to assess and annotate those prespecified constructs.

LLMs for clinical knowledge and causal-graph elicitation. Medical LLMs encode substantial clinical knowledge, but their outputs require verification (Singhal et al. 2023). Published causal-discovery work uses node metadata or observational data to prompt LLMs to construct graphs, while systematic evaluations show that causal judgments can depend on memorization, incorrect relationships in pretraining data, and contextual wording (Roy et al. 2025; Feng et al. 2025). Earlier medical work constructed causal graphs by combining relationships extracted from clinical literature with associations observed in a large EMR repository and evaluating the resulting edges with medical experts (Nordon et al. 2019). These approaches begin with a predefined variable set, use patient-level observational evidence, or both. CliniCause instead withholds patient records from the LLM and uses schema-level column names and literature-grounded clinical knowledge to elicit the proxy variables, their operational cutoffs and temporal rules, and then their DAG.

Causal estimators and their evaluation. Double machine learning uses orthogonal scores and cross-fitting to accommodate flexible nuisance estimation, while causal and generalized random forests model heterogeneous effects (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019). CausalPFN amortizes effect estimation through an in-context model trained across simulated data-generating processes (Meresht et al. 2025). Causal estimators are commonly evaluated using semi-synthetic benchmarks such as IHDP and ACIC, experimentally anchored constructions such as Jobs/LaLonde and OSRCT, or empirically anchored generators. Influence-function validation, observational sampling from randomized trials, and Credence make such evaluations substantially more realistic and diverse (Alaa and van der Schaar 2019; Gentzel, Pruthi, and Jensen 2021; Parikh et al. 2022). Nevertheless, these frameworks evaluate estimators against experimental or constructed targets; they do not provide the knowledge-grounded observational representation contributed here.

Causal analysis of observational clinical records. Causal analyses of EHR data require explicit treatment strategies, time zero, eligibility criteria, outcomes, and adjustment assumptions. Target-trial emulation and work on well-defined interventions formalize these requirements and expose the ambiguity of treating observational clinical states as interventions (Hernan and Robins 2016; Hernan and Taubman 2008). Reviews of individualized treatment-effect estimation from patient records further emphasize challenges in confounding, missingness, measurement, and treatment definition (Bica et al. 2021). Such studies generally reconstruct a cohort and causal representation for one clinical question. CliniCause instead packages multiple clinically motivated candidate exposures within a shared interface for cohort definitions, proxy rules, DAGs, adjustment sets, estimators, and provenance.

Positioning. Prior work has separately developed knowledge-based phenotyping, LLM-assisted causal graphs, causal estimators, and study-specific analyses of observational clinical records. CliniCause connects these lines at a different boundary: an LLM with no patient-level access externalizes literature-grounded clinical knowledge as operational proxy definitions and a causal graph; deterministic code applies that representation to source-observed ICU records; and the resulting resources are examined through proxy recoverability, cross-estimator agreement, and separate clinical-literature corroboration. We are not aware of prior published work combining this complete sequence within a reusable observational causal-analysis resource.

3 Methodology

In this section, we describe the CliniCause construction pipeline summarized in Figure 1: resource definition, knowledge-driven proxy and graph design, patient-level instantiation, and provenance tracking. The subsequent clinical-evidence validation is described in detail in Section 5.3.

3.1 Clinical Setting and Resource Contract

Let $d \in \{\mathrm{MIMIC}, \mathrm{PhysioNet}\}$ index the source database and let

$$\mathcal{I}_d = \{1, \ldots, N_d\}$$

denote its set of analysis records. An analysis record is an ICU stay or source time-series record, not necessarily a unique patient. For record $i \in \mathcal{I}_d$, the irregular clinical observations are

$$\mathcal{X}_i^{(d)} = \{(t_{ij}, v_{ij}, x_{ij})\}_{j=1}^{n_i},$$

where t_{ij}, v_{ij}, and x_{ij} denote the observation time, measured variable, and recorded value. A linked record-level view supplies the canonical identifier $\mathrm{id}_i^{(d)}$, baseline covariates $\mathbf{C}_i^{(d)}$, and source-observed in-hospital mortality outcome $Y_i^{(d)}$. We write the complete source input as

$$\mathcal{D}_d^{\mathrm{src}} = \left\{ \left(\mathrm{id}_i^{(d)},\, \mathcal{X}_i^{(d)},\, \mathbf{C}_i^{(d)},\, Y_i^{(d)} \right) :\; i \in \mathcal{I}_d \right\}.$$

MIMIC-III and PhysioNet 2012 are processed independently (Johnson et al. 2016; Silva et al. 2012); their records, proxy semantics, graphs, and estimates are never pooled.

CliniCause is a factored transformation with separate knowledge-design and patient-instantiation stages. Let $\mathcal{S}_d$ denote the source schema available to the design stage and let $\mathcal{L}$ denote external clinical knowledge and literature. The knowledge function

$$\mathcal{K}_d : \left(\mathcal{S}_d,\, Y^{(d)},\, \mathcal{L}\right) \longmapsto (\mathcal{Q}_d,\, G_d)$$

constructs a set of operational proxy definitions $\mathcal{Q}_d$ and a dataset-specific DAG G_d. The LLM participates only in $\mathcal{K}_d$ and has no access to $\mathcal{D}_d^{\mathrm{src}}$.

A deterministic instantiation function $\mathcal{F}_d$ then applies the accepted proxy definitions and DAG to $\mathcal{D}_d^{\mathrm{src}}$. This separation ensures that schema-level knowledge elicitation occurs without patient-level access, while patient records enter only during deterministic instantiation. The resulting causal-analysis resource is

$$\mathcal{R}_d = \left(\mathcal{I}_d,\, \widetilde{\mathbf{Z}}^{(d)},\, \mathbf{Y}^{(d)},\, \mathbf{C}^{(d)},\, G_d,\, \{\mathcal{A}_k^{(d)}\}_{k=1}^{K_d},\, \mathrm{Prov}_d\right),$$

where $\widetilde{\mathbf{Z}}^{(d)}$ contains the instantiated proxy exposures, $\mathbf{Y}^{(d)}$ contains the source-observed outcomes, $\mathbf{C}^{(d)}$ contains the baseline covariates, $\mathcal{A}_k^{(d)}$ is the recorded adjustment set for exposure k, and Prov_d contains configuration, artifact, and lineage metadata. Thus, the irregular time series $\{\mathcal{X}_i^{(d)}\}_{i\in\mathcal{I}_d}$ are source inputs, whereas $\widetilde{\mathbf{Z}}^{(d)}$ is the constructed tabular exposure interface.

The shared resource contract standardizes analytical roles rather than imposing a common clinical vocabulary. The admitted resources contain 26,845 MIMIC-III records with $K_{\mathrm{MIMIC}} = 9$ proxy exposures and 7,993 PhysioNet records with $K_{\mathrm{PhysioNet}} = 10$.

3.2 Knowledge-Driven Proxy and Graph Design

The knowledge-driven design stage first defines proxy variables and their operational rules, and only then organizes those constructs into a causal DAG. For source d, the LLM receives the available column names $\mathcal{S}_d$ and target outcome Y, but no patient rows, empirical distributions, exposure prevalences, or effect estimates. The elicitation step was conducted interactively through the ChatGPT desktop interface using the prompt reproduced in Supplement **??**. This is demonstrated in Figure 2

Dan: Kobi - prepare supplementary PDF with all ChatGPT prompts - to generation variables, DAG, etc

. Abstractly, it is represented as

$$\mathcal{K}_d : (\mathcal{S}_d, Y^{(d)}) \longmapsto \left\{ \left(Z_k^{(d)}, r_k^{(d)}, \mathcal{L}_k^{(d)}\right) \right\}_{k=1}^{K_d},$$

where $Z_k^{(d)}$ is a candidate clinical state, $r_k^{(d)}$ contains its cutoffs and temporal rules, and $\mathcal{L}_k^{(d)}$ records its clinical rationale and supporting literature. ChatGPT was instructed to ground every proposed proxy and cutoff in published clinical evidence. The authors inspected the proposed rationales and

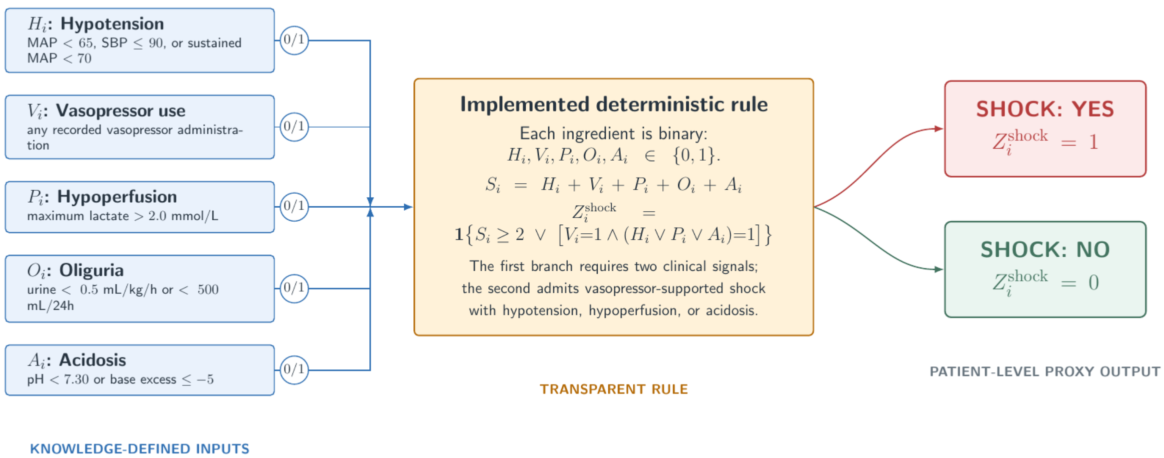

Figure 2: Example of ChatGPT-assisted, knowledge-driven proxy construction for the shock exposure. Five binary clinical indicators (H_i, V_i, P_i, O_i, A_i) are derived from the corresponding patient time series and combined by an explicit deterministic rule to produce the patient-level proxy Z_i^{shock}. ChatGPT proposes candidate indicators, cutoffs, and combination logic at design time; the authors inspect and encode the accepted specification.

translated the selected definitions into deterministic source code. Because ChatGPT received the schema and target outcome but neither patient records nor empirical effect estimates, this stage could not select proxy definitions according to their observed MIMIC-III or PhysioNet effects. After fixing the proxy set, the LLM proposes a directed graph

$$G_d = (V_d, E_d), \qquad V_d = \mathbf{C}^{(d)} \cup \mathbf{Z}^{(d)} \cup \{Y^{(d)}\},$$

with a clinical rationale and supporting source required for each proposed edge $(u \to v) \in E_d$. The authors inspect the proposed relationships and encode the selected graph.

3.3 Patient-Level Instantiation and Resource Assembly

The accepted proxy definitions are first applied to each patient's irregular time series using deterministic code. Consider shock as a running example. Its definition searches the available record for prespecified clinical evidence, such as hypotension, elevated lactate, vasopressor administration, or low urine output, using dataset-specific measurements and temporal rules. If at least one required condition is observed, the rule-based shock proxy is positive. Missing measurements provide no positive evidence and are not silently interpreted as normal values. The same procedure is applied to every proxy. These labels are operational research constructs, not chart-adjudicated diagnoses or literal clinical interventions.

The deterministic rules define the clinical meaning of each proxy, but the archived CliniCause resources add a model-assisted annotation step. STraTS, GRU-D, GRU, and TCN are trained to recover the complete vector of rule-derived proxy labels from the irregular time series (Tipirneni and Reddy 2022; Che et al. 2018; Cho et al. 2014; Bai, Kolter, and Koltun 2018). The models never receive mortality as a target and do not select proxy names, measurements, cutoffs, temporal rules, or DAG edges. Their role is therefore to provide alternative patient-level annotations of a knowledge-defined construct, not to discover the construct itself.

For each patient i and proxy k, the final archived exposure is the fixed majority vote of the deterministic rule and the four model predictions:

$$\widetilde{Z}_{ik}^{(d)} = \mathbb{1}\left\{ Z_{ik}^{(d,\text{rule})} + \sum_{m=1}^{4} \widehat{Z}_{ik}^{(d,m)} \geq 3 \right\}.$$

A record is admitted only when all five annotations are present, preventing a missing model output from changing the voting denominator. Because the models can alter a patient-level rule label through this vote, the resulting exposures are most precisely described as *knowledge-defined and model-assisted*. Predictive evaluation, reported separately, measures how well the models recover the rule targets; it does not establish the clinical correctness of those targets.

The resulting estimator-agnostic resource row is

$$R_i^{(d)} = \left(\mathrm{id}_i^{(d)}, \widetilde{\mathbf{Z}}_i^{(d)}, Y_i^{(d)}, \mathbf{C}_i^{(d)} \right).$$

The resource stores the full proxy vector once. For a causal analysis of proxy k, $\widetilde{Z}_{ik}^{(d)}$ becomes the exposure, in-hospital mortality remains the outcome, and the corresponding graph-derived adjustment variables are selected from the observed covariate and non-treatment proxy fields. Recording these exposure-specific adjustment sets makes the causal-analysis interface inspectable and replaceable without reconstructing the patient cohort.

4 Empirical Evaluation

Without known causal ground truth, CliniCause requires a converging-evidence evaluation scheme. This section describes its three pillars: proxy recoverability, cross-estimator agreement, and clinical-literature corroboration.

Resource	Records	Proxies	Outcome
MIMIC-III	26,845	9	In-hospital mortality
PhysioNet 2012	7,993	10	In-hospital mortality

Table 1: Primary original-cohort CliniCause resources. Counts refer to analysis records and admitted non-chronic proxy exposures.

4.1 Predictive Recoverability

We first ask whether the knowledge-based proxy rules are supported by patterns in the observed ICU time series. The accepted ChatGPT-assisted rules assign each proxy a binary value. This yields nine binary prediction targets in MIMIC-III and ten in PhysioNet 2012, one for each admitted exposure. We train four models designed for irregular time series—STraTS, GRU, GRU-D, and TCN—to predict these rule-derived labels from the underlying patient measurements (Tipirneni and Reddy 2022; Cho et al. 2014; Che et al. 2018; Bai, Kolter, and Koltun 2018).

We report macro AUROC, AUPRC over a 3-fold CV test.

Dan: verify

4.2 Effect Estimation and Stability Metrics

We evaluate all 19 admitted dataset–exposure comparisons: nine in MIMIC-III and ten in PhysioNet. For each comparison, one binary proxy Z_k is designated as the exposure T_k, and source-observed in-hospital mortality is the outcome Y. The datasets are analyzed separately because their cohorts, proxy definitions, and DAGs differ. All prespecified comparisons are retained, including weak, negative, and estimator-discordant estimates.

We apply three estimators. CausalForestDML is the primary nonlinear estimator and combines orthogonalized nuisance estimation with a causal-forest final stage (Chernozhukov et al. 2018; Wager and Athey 2018; Athey, Tibshirani, and Wager 2019; Oprescu et al. 2019). LinearDML uses the same DML framework with a linear final effect model. CausalPFN provides a distinct pretrained estimation paradigm (Meresht et al. 2025).

For each exposure, the dataset-specific DAG determines the observed variables used for confounding adjustment; remaining proxies are not included automatically. The DML estimators receive adjustment variables and effect-modification variables through separate inputs, whereas CausalPFN receives their deduplicated union. Thus, the estimators use the same exposure, outcome, cohort, and available covariate information, while processing that information differently.

For estimator e, exposure k, and dataset d, we summarize the fitted record-level conditional effects by

$$\widehat{\mu}_{ek}^{(d)} = \frac{1}{n_d}\sum_{i=1}^{n_d}\widehat{\tau}_{ek}^{(d)}(i),$$

which we call the *mean model-estimated CATE*. We assess cross-estimator stability in three ways. First, we count exposure comparisons for which all estimators agree in sign. Second, we calculate pairwise Spearman correlations between the exposure rankings produced by the estimators within each dataset. Third, for each estimator pair (e, e'), we calculate

$$\mathrm{RMSE}_{e,e'}^{(d)} = \sqrt{\frac{1}{K_d}\sum_{k=1}^{K_d}\left(\widehat{\mu}_{ek}^{(d)} - \widehat{\mu}_{e'k}^{(d)}\right)^2},$$

and report the range across estimator pairs. This RMSE measures numerical disagreement between estimators, not error relative to causal ground truth. We also inspect the per-exposure range across the three estimates to identify constructs whose conclusions are especially estimator sensitive.

Dan: How about matching? AIPW?

4.3 Clinical-Evidence Corroboration

After resource construction and causal estimation, we conducted a separate literature audit to assess whether the estimated effects were broadly compatible with mortality patterns reported for corresponding clinical constructs. For each CliniCause proxy, we searched for ICU studies examining the nearest clinically recognized state and reporting a numerical mortality contrast. For example, renal dysfunction was mapped to acute kidney injury defined using creatinine or urine-output criteria, while shock was mapped to comparisons of septic shock with sepsis without shock. These mappings are construct-level analogies; they do not imply that the published exposure and the CliniCause proxy are identical.

Candidate studies were identified through parallel searches in NotebookLM, Perplexity, and alphaXiv using the common prompt reported in Section XX of the supplementary material. We prioritized explicit causal analyses, followed by matched or adjusted observational comparisons and, when no stronger comparator was available, unadjusted mortality contrasts. Numerical comparisons were made only when absolute mortality differences were reported or directly recoverable; odds ratios and hazard ratios were retained on their original scales. Agreements and discrepancies were recorded symmetrically. Because the studies differ in construct definition, population, adjustment strategy, and outcome horizon, their results are treated as contextual clinical evidence rather than causal ground truth or calibration targets.

5 Results

In this section, we report the results of the empirical evaluation protocol described in Section 4. Table 1 summarizes the two CliniCause resources included in the evaluation. All experiments were conducted using **[HARDWARE STUB: CPU, GPU, memory, and software environment]**.

Dan: Describe the hardware and code used to run the experiments - Python code, who wrote it? did we use existing git-repos of the Causal estiamtors?

5.1 Construct Plausibility and Predictive Recoverability

The informal ICU consultations found the selected constructs and graph relationships broadly plausible, but produced no recorded ratings or independent adjudication statistics.

We next evaluate whether irregular-time-series models can predict the ChatGPT-derived proxy values. Table 2 shows consistently high held-out performance across datasets and architectures, with AUROC values of 0.867–0.918 and AUPRC values of 0.823–0.905. These results indicate that the knowledge-defined proxy rules and cutoffs correspond to patterns recoverable from the observed ICU time series, although we do not claim that they establish clinical or causal validity.

Dataset	Model	AUROC	AUPRC
MIMIC-III	STraTS	**0.905**	**0.869**
	GRU	0.881	0.840
	GRU-D	0.884	0.841
	TCN	0.867	0.823
PhysioNet	STraTS	0.915	0.877
	GRU	0.915	0.897
	GRU-D	**0.918**	**0.905**
	TCN	0.899	0.875

Table 2: 3-CV Held-out prediction of rule-derived proxy targets. Values are macro-averaged. Bold indicates the highest value within each dataset

5.2 Cross-Estimator Stability of Effect Patterns

We next examine whether the estimated effect patterns remain stable across the three causal estimators. The main text summarizes agreement in direction, ranking, and magnitude, while the complete exposure-level estimator matrix is reported in Section XX of the supplementary material. Overall, the three estimators produced highly consistent effect patterns (Table 3).

CausalForestDML and LinearDML agreed in direction in all 19 dataset–exposure comparisons, while all three estimators agreed in 18 of 19. The sole sign disagreement was PhysioNet shock, for which the two DML estimates were negative and the CausalPFN estimate was slightly positive.

Within-dataset rankings were particularly stable: pairwise Spearman correlations ranged from 0.983 to 1.000 in MIMIC-III and from 0.794 to 0.964 in PhysioNet. Pairwise RMSE remained between 1.08 and 2.57 percentage points. The largest exposure-level ranges were observed for MIMIC-III cardiac strain (7.17 pp), PhysioNet hepatic dysfunction (4.67 pp), PhysioNet shock (3.11 pp), and PhysioNet renal dysfunction (2.94 pp).

As a descriptive cross-dataset comparison, cardiac strain had the largest three-estimator average in both resources, and eight of nine approximately shared constructs had the same effect direction. Their average-effect rankings were only moderately correlated ($\rho = 0.533$), however, reflecting differences in cohorts, proxy definitions, measurements, DAGs, and adjustment sets. The results therefore support stability across estimators once a CliniCause representation is fixed, rather than equivalence of the two resources or correctness of the estimated causal effects. Full exposure-level estimates are reported in the supplementary material.

Dataset	Sign agreement	Spearman ρ	Pairwise RMSE
MIMIC-III	9/9	0.983–1.000	1.35–2.57 pp
PhysioNet 2012	9/10	0.794–0.964	1.08–2.04 pp

Table 3: Cross-estimator stability across CausalForestDML, LinearDML, and CausalPFN. Sign agreement requires all three estimators to agree. Spearman correlations and RMSE values give the ranges across the three estimator pairs. RMSE measures estimator disagreement, not error relative to causal ground truth.

Comparison pattern	Clinical constructs
Similar direction and scale	Renal/AKI; hepatic/bilirubin; cardiac injury
Dataset dependent	Inflammation/sepsis: close in PhysioNet, larger in MIMIC-III
Substantial scale or directional discrepancy	Shock; coagulation; respiratory; neurologic; metabolic
No comparable numerical contrast	Global severity

Table 4: Qualitative synthesis of the clinical-evidence comparison. Categories summarize approximate direction and scale; the clinical studies do not provide ground-truth CATEs. Full numerical comparisons, study designs, mortality horizons, and citations are reported in the supplement.

5.3 Clinical-Evidence Corroboration

We applied the clinical-evidence validation protocol described in Section 4.3. Table 4 summarizes the comparison, with the complete numerical evidence audit reported in the supplement. The renal comparison provides the closest match: a published attributable 30-day mortality estimate of 8.1 percentage points is close to the CliniCause ranges of 8.7–9.1 in MIMIC-III and 9.1–12.0 in PhysioNet. Hepatic and cardiac estimates were also similar in direction and approximate scale, while inflammation/sepsis aligned more closely in PhysioNet than in MIMIC-III.

Shock was the clearest discrepancy. The published mortality contrast was 11.1 points, compared with 2.1–3.2 in MIMIC-III and -2.7–0.4 in PhysioNet. Coagulation, respiratory, neurologic, and metabolic estimates also differed substantially from their published comparators. These differences may reflect construct severity, proxy thresholds, temporal windows, population and outcome differences, adjustment choices, or estimator behavior. They identify priorities for targeted sensitivity analyses rather than grounds for selecting or discarding results.

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6 Discussion and Limitations

CliniCause demonstrates a representation-centered alternative to constructing causal benchmarks. Rather than simulate exposure assignments or outcomes, the pipeline preserves source-observed clinical processes and places researcher intervention in an inspectable layer of proxy definitions, temporal rules, causal assumptions, and adjustment choices. This approach is not limited to ICU records. Public longitudinal data in education, transportation, environmental monitoring, energy systems, and industrial telemetry may offer similar opportunities wherever domain knowledge can map raw measurements to meaningful candidate exposures and outcomes.

A natural next step is to turn these resources into an expanding evaluation system for published causal methods. The small number of widely used benchmarks currently makes it possible for method development to adapt, directly or indirectly, to much of the available evaluation suite. New resources could instead be added continually and used to stress-test methods across previously unseen populations, representations, support conditions, and assumption violations. Patient-level test data and outcome summaries could remain hidden, with submitted estimators evaluated through a controlled interface. Such an infrastructure would reveal where methods generalize, disagree, or fail without allowing developers to tune directly to every evaluation dataset.

CliniCause complements rather than replaces existing benchmarks. Synthetic and semi-synthetic designs provide exact effects under specified data-generating mechanisms; RCT-derived evaluations provide experimental anchors but depend on sampling and transport assumptions; and data-fusion benchmarks depend on the comparability of their source populations. These resources make accuracy measurable, but performance remains conditional on researcher-specified data-generating, sampling, or data-fusion mechanisms. CliniCause makes the opposite tradeoff: it gives up a known answer key in exchange for evaluation under source-observed measurement, support, missingness, and outcome processes.

Several limitations bound the present contribution. The proxy states are operational analytical constructs rather than chart-adjudicated phenotypes or literal interventions, and physician feedback was informal rather than a documented independent review. The LLM-assisted design was not compared with clinician-only, literature-only, or alternative-LLM construction. MIMIC-III and PhysioNet access conditions restrict redistribution and complicate independent reproduction. Finally, estimator agreement, predictive recoverability, and clinical-literature corroboration provide converging but non-identifying evidence. They do not establish proxy validity, graph correctness, causal identification, or external clinical validity, and the resulting estimates must not be interpreted as treatment recommendations. This limitation is not unique to CliniCause: synthetic and semi-synthetic benchmarks provide causal truth within their specified mechanisms, but whether that constructed truth reflects causal processes operating in nature remains an open question for the field.

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